Linear Algebra I — Week 5

Bases and Dimension

Name: _____ Date: _____ All work **by hand**: no calculator.

Key Facts

Basis. A subset S of a vector space V is a *basis* if it is linearly independent and spans V. Equivalently, every vector of V can be written *uniquely* as a linear combination of the vectors of S (spanning gives existence, independence gives uniqueness). Independence and dimension depend on the field F of scalars ($\mathbb{C}^2$ has dimension 2 over $\mathbb{C}$ but 4 over $\mathbb{R}$).
Dimension. V is *finite-dimensional* if it has a finite basis; then every basis has the same number of elements, called $\dim V$. Standard bases: $\{e_1, \ldots, e_n\}$ for F^n ($\dim n$); $\{E_{ij}\}$ for $M_{m\times n}(F)$ ($\dim mn$); $\{1, x, \ldots, x^n\}$ for $P_n(F)$ ($\dim n+1$). $P(\mathbb{R})$, all polynomials, is not finite-dimensional.
Dimension Bound. If V is spanned by n vectors, every linearly independent set in V has at most n elements. So in a space with $\dim V = n$: more than n vectors are linearly dependent; fewer than n vectors cannot span.
Extension Lemma. If S is linearly independent and $x \notin S$, then $S \cup \{x\}$ is linearly independent iff $x \notin \operatorname{span}(S)$. A linearly independent set can be extended to a basis by adding vectors from a spanning set one at a time, each outside the current span; a finite spanning set can be shrunk to a basis by discarding vectors lying in the span of the others.
Basis criterion. If $\dim V = n$ and S has n elements, then S is a basis iff S is linearly independent iff S spans V. If W is a subspace of V then $\dim W \leq \dim V$, with equality only if $W = V$.
Computing. Translate vectors, polynomials and matrices into coordinate vectors with respect to a standard basis, then row reduce. The pivot columns of the *original* matrix are a basis of the span of the columns; each non-pivot column of the RREF gives the coefficients of that column in terms of the pivot columns. A subspace defined by linear conditions on the coordinates is the solution set of a homogeneous system: its dimension is the number of free variables, and the vectors multiplying the free parameters form a basis.

Worked Examples

Example 1 · Standard bases and dimensions

Prove that $\{1, x, x^2\}$ is a basis of $P_2(F)$ and that $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ is a basis of $M_2(F)$, for any field F. State the dimensions.

Solution.

1. $P_2(F)$ consists of the polynomials $a+bx+cx^2$ with $a, b, c \in F$. *Spanning:* each such polynomial is $a \cdot 1 + b \cdot x + c \cdot x^2$, a combination of $1, x, x^2$.
2. *Independence:* if $a\cdot 1+b\,x+c\,x^2$ is the zero polynomial, all its coefficients are 0, so $a = b = c = 0$. Hence $\{1, x, x^2\}$ is a basis and $\dim P_2(F) = 3$. In general $\dim P_n(F) = n+1$.
3. $M_2(F)$: $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = aE_{11} + bE_{12} + cE_{21} + dE_{22}$, so the four matrices span. If $aE_{11} + bE_{12} + cE_{21} + dE_{22} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is the zero matrix, then $a = b = c = d = 0$, so they are independent.
4. So $\dim M_2(F) = 4$. In general $\dim M_{m\times n}(F) = mn$, $\dim F^n = n$. None of these dimensions depends on which field F is used (as long as the scalars are F).

Answer: $\dim P_2(F) = 3$ with basis $\{1, x, x^2\}$; $\dim M_2(F) = 4$ with basis $\{E_{ij}\}$.

Example 2 · Is a set a basis of $\mathbb{R}^3$?

Decide whether $\{(1,2,0),(0,1,1),(1,0,-2)\}$ is a basis of $\mathbb{R}^3$. If not, describe what it spans.

Solution.

1. Put the vectors in the columns: $A = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & -2 \end{pmatrix}$. Reduce.
2. Apply $R_2 \to R_2 - 2R_1$:
$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{pmatrix}$$
3. Apply $R_3 \to R_3 - R_2$:
$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$
4. Pivots are in columns 1 and 2 only; column 3 has no pivot, so the vectors are linearly dependent. Column 3 of the RREF is $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$, so $v_3 = 1\cdot v_1 - 2\cdot v_2$. Check: $(1,2,0) - 2(0,1,1) = (1,0,-2)$ ✓.
5. Three vectors that are dependent cannot be a basis. Since v_1, v_2 are independent (pivot columns) and span the same space, the set spans a 2-dimensional subspace (a plane) of $\mathbb{R}^3$, with basis $\{(1,2,0),(0,1,1)\}$.

Answer: Not a basis: $v_3 = v_1 - 2v_2$. The span is a plane with basis $\{(1,2,0),(0,1,1)\}$.

Example 3 · Basis of a constraint subspace

Find a basis and the dimension of $W = \{p \in P_3(\mathbb{R}) :\ p(1) = 0,\ p(-1) = 0\}$.

Solution.

1. Write $p(x) = a + bx + cx^2 + dx^3$, coordinates (a,b,c,d). Then $p(1) = a + b + c + d$ and $p(-1) = a - b + c - d$. The conditions give the homogeneous system $\begin{cases} a+b+c+d = 0 \\ a-b+c-d = 0 \end{cases}$.
2. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:
$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & -1 & 0 \end{array}\right)$$
3. Apply $R_2 \to R_2 - R_1$:
$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & -2 & 0 & -2 & 0 \end{array}\right)$$
4. Apply $R_2 \to \dfrac{1}{-2} R_2$:
$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array}\right)$$
5. Apply $R_1 \to R_1 - R_2$:
$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array}\right)$$

6. Pivot columns are the columns of the basic variables: a, b. The remaining variable(s) c, d are *free*: call them $c = s$, $d = t$.
7. Read each nonzero row as an equation for its basic variable:

   $a = -s$

   $b = -t$

   $c = s$

   $d = t$
8. Write this as a vector: $\mathbf{x} = s\begin{pmatrix}-1\\0\\1\\0\end{pmatrix} + t\begin{pmatrix}0\\-1\\0\\1\end{pmatrix}$ (the parameters s, t may be any numbers).
9. Check with $s = 1$, $t = 1$: $\mathbf{x} = \begin{pmatrix}-1\\-1\\1\\1\end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(0, 0)$ ✓.
10. Translate each coordinate vector back into the original objects. Setting one parameter to 1 and the others to 0 in the vector form gives: $\begin{pmatrix}-1\\0\\1\\0\end{pmatrix} \leftrightarrow -1 + x^2$; $\begin{pmatrix}0\\-1\\0\\1\end{pmatrix} \leftrightarrow -x + x^3$.
11. These 2 vectors *span* W: the vector form says every element of W equals $s\,\mathbf{b}_1 + t\,\mathbf{b}_2$ with the parameters free. They are *linearly independent*: in this combination the entry in the position of the free variables c, d equals the parameters s, t (each basis vector has 1 in its own free position and 0 in the other free positions), so a combination equal to $\mathbf{0}$ forces every parameter to be 0.
12. Hence the objects above form a basis of W, and $\dim W$ equals the number of free variables, namely 2.
13. Check: $q = -1 + x^2$ has $q(1) = 0$ and $q(-1) = 0$ ✓; $r = -x + x^3$ has $r(1) = 0$ and $r(-1) = 0$ ✓.

Answer: Basis: $\{-1 + x^2,\ -x + x^3\}$; $\dim W = 2$.

Example 4 · Extracting a basis from a spanning set

The set $S = \{(1, 1, 0), (0, 1, 1), (1, 2, 1), (1, 0, -1)\}$ spans a subspace U of $\mathbb{R}^3$. Find a basis of U chosen from S, and write the other vectors through it.

Solution.

1. The columns of $A = \begin{pmatrix}1 & 0 & 1 & 1\\1 & 1 & 2 & 0\\0 & 1 & 1 & -1\end{pmatrix}$ are the vectors of S. Reduce.
2. Apply $R_2 \to R_2 - R_1$:

   $$\begin{pmatrix}1 & 0 & 1 & 1\\0 & 1 & 1 & -1\\0 & 1 & 1 & -1\end{pmatrix}$$
3. Apply $R_3 \to R_3 - R_2$:

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

4. Pivot columns 1 and 2. Rule: the corresponding columns of the *original* matrix form a basis of the span. So $\{(1,1,0),(0,1,1)\}$ is a basis of U and $\dim U = 2$.
5. Column 3 of the RREF is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$: $v_3 = v_1 + v_2$; column 4 is $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$: $v_4 = v_1 - v_2$. Check: $(1,1,0)+(0,1,1) = (1,2,1)$ ✓and $(1,1,0)-(0,1,1) = (1,0,-1)$ ✓.

Answer: Basis $\{(1,1,0),(0,1,1)\}$, $\dim U = 2$; $v_3 = v_1 + v_2$, $v_4 = v_1 - v_2$.

Example 5 · Extending an independent set to a basis

Extend the linearly independent set $\{(1,1,1),(1,2,3)\}$ to a basis of $\mathbb{R}^3$ using standard basis vectors.

Solution.

1. Put v_1, v_2 followed by e_1, e_2, e_3 in the columns and reduce; keep the pivot columns (Extension Lemma: each kept vector is outside the span of those before it). $B = \left(\begin{array}{cc|ccc} 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 0 & 1 \end{array}\right)$.
2. Apply $R_2 \to R_2 - R_1,\ R_3 \to R_3 - R_1$:

$$\left(\begin{array}{cc|ccc} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 2 & -1 & 0 & 1 \end{array}\right)$$

3. Apply $R_1 \to R_1 - R_2,\ R_3 \to R_3 - 2R_2$:

$$\left(\begin{array}{cc|ccc} 1 & 0 & 2 & -1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 \end{array}\right)$$

4. Apply $R_1 \to R_1 - 2R_3,\ R_2 \to R_2 + R_3$:

$$\left(\begin{array}{cc|ccc} 1 & 0 & 0 & 3 & -2 \\ 0 & 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & -2 & 1 \end{array}\right)$$

5. The pivot columns are $1, 2, 3$. Column 3 is e_1, so $\{(1,1,1),(1,2,3),(1,0,0)\}$ is a basis (three independent vectors in $\mathbb{R}^3$). Columns 4 and 5 (e_2, e_3) have no pivot: each is already in the span of the previous vectors.
6. Direct check that $e_1 \notin \operatorname{span}\{v_1, v_2\}$: $a(1,1,1) + b(1,2,3) = (1,0,0)$ needs $a + 2b = 0$ and $a + 3b = 0$, so $b = 0$, $a = 0$, but then the first coordinate is $0 \neq 1$. Not in the span ✓.

Answer: $\{(1,1,1),(1,2,3),(1,0,0)\}$ is a basis of $\mathbb{R}^3$.

Example 6 · Coordinates in $\mathbb{C}^2$

Show that $\beta = \{(1,i),(1,-i)\}$ is a basis of $\mathbb{C}^2$ over $\mathbb{C}$ and find $[w]_\beta$ for $w = (3, 1+i)$.

Solution.

1. We need $a(1,i) + b(1,-i) = (3, 1+i)$, i.e. $a + b = 3$ and $ia - ib = 1 + i$. The matrix has the basis vectors as columns.
2. Write the system as an augmented matrix, one row per equation, with a bar before the

right-hand side:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ i & -i & 1+i \end{array}\right)$$

3. Apply $R_2 \to R_2 - iR_1$:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -2i & 1-2i \end{array}\right)$$

4. Apply $R_2 \to \dfrac{1}{-2i}R_2$:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1+\frac{1}{2}i \end{array}\right)$$

5. Apply $R_1 \to R_1 - R_2$:

$$\left(\begin{array}{cc|c} 1 & 0 & 2-\frac{1}{2}i \\ 0 & 1 & 1+\frac{1}{2}i \end{array}\right)$$

6. Every column of the coefficient part holds a pivot, so there are no free variables and the solution is unique.

7. Read each nonzero row as an equation for its basic variable:

$$a = 2 - \frac{1}{2}i$$
$$b = 1 + \frac{1}{2}i$$

8. Check by substituting $(a,b) = (2-\frac{1}{2}i, 1+\frac{1}{2}i)$ into the original equations: the left sides give $(3, 1+i)$, the right-hand sides ✓.

9. The coefficient part reduced to I_2 (a pivot in every row and column), so for every w there is exactly one solution: β spans and is independent, hence is a basis, and coordinates are unique.

10. So $[w]_\beta = (2-\frac{1}{2}i,\ 1+\frac{1}{2}i)^T$. Check: $a+b=3$ ✓and $i(a-b) = i(1-i) = i+1$ ✓.

Answer: β is a basis and $[w]_\beta = \begin{pmatrix} 2-\frac{1}{2}i \\ 1+\frac{1}{2}i \end{pmatrix}$.

Example 7 · Counting arguments

Decide using dimensions only: (a) can $\{1, x\}$ span $P_2(\mathbb{R})$? (b) are three vectors in $\mathbb{C}^2$ (over $\mathbb{C}$) necessarily dependent? (c) can a set of 6 vectors be a basis of $M_{2\times 3}(\mathbb{R})$? (d) can 4 independent vectors lie in a subspace of $\mathbb{R}^5$ of dimension 3?

Solution.

1. (a) $\dim P_2(\mathbb{R}) = 3$ and a spanning set has at least 3 vectors. $\{1, x\}$ has 2, so it cannot span.
2. (b) $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$, so $\mathbb{C}^2$ is spanned by 2 vectors and every independent set has at most 2 elements. So any 3 vectors are dependent: yes.
3. (c) $\dim M_{2\times 3}(\mathbb{R}) = 6$, so a basis has exactly 6 elements. A set of 6 vectors *can* be a basis (it is a basis iff it is independent, or iff it spans); the count alone does not decide it.
4. (d) A subspace U with $\dim U = 3$ is spanned by 3 vectors, so every independent set in U has at most 3 elements. 4 independent vectors cannot lie in U: no.

Answer: (a) No. (b) Yes. (c) Possibly (count is consistent; check independence). (d) No.

Example 8 · Proof: the Extension Lemma

Let S be linearly independent in V and $x \in V$ with $x \notin S$. Prove: $S \cup \{x\}$ is linearly independent if and only if $x \notin \operatorname{span}(S)$.

Solution.

1. ($\Rightarrow$) We prove the contrapositive. Suppose $x \in \operatorname{span}(S)$: $x = c_1 s_1 + \cdots + c_m s_m$ with $s_i \in S$. Then $1 \cdot x - c_1 s_1 - \cdots - c_m s_m = \mathbf{0}$ is a relation among distinct elements of $S \cup \{x\}$ with coefficient $1 \neq 0$ on x. So $S \cup \{x\}$ is dependent.
2. ($\Leftarrow$) Suppose $x \notin \operatorname{span}(S)$ and $\alpha x + c_1 s_1 + \cdots + c_m s_m = \mathbf{0}$ with $s_i \in S$ distinct. If $\alpha \neq 0$, divide by α (possible in a field): $x = -\alpha^{-1}(c_1 s_1 + \cdots + c_m s_m) \in \operatorname{span}(S)$, contradicting $x \notin \operatorname{span}(S)$.
3. So $\alpha = 0$, leaving $c_1 s_1 + \cdots + c_m s_m = \mathbf{0}$. Since S is independent, every $c_i = 0$. All coefficients are 0, so $S \cup \{x\}$ is independent.

Answer: A relation involving x with $\alpha \neq 0$ would put x in $\operatorname{span} S$; if $x \notin \operatorname{span} S$ then $\alpha = 0$ and independence of S kills the rest.

Practice Problems

Part A — Standard Bases, Counting and Dimension

1. [Foundational]
State the standard basis and the dimension of each space: (a) $\mathbb{C}^3$ over $\mathbb{C}$; (b) $M_{2\times 3}(\mathbb{R})$; (c) $P_3(\mathbb{R})$. Then write $A = \begin{pmatrix} 2 & -1 & 0 \\ 5 & 3 & 4 \end{pmatrix}$ and $p(x) = 4 - x + 2x^3$ as combinations of the standard bases of (b) and (c).

2. [Foundational]
For each statement decide whether it is true, using only a dimension count (state the dimension used and the theorem): (a) five matrices in $M_2(\mathbb{R})$ must be linearly dependent; (b) $\{(1,0,0),(0,1,0)\}$ spans $\mathbb{R}^3$; (c) any four polynomials in $P_2(\mathbb{R})$ are linearly dependent; (d) five matrices in $M_{3\times 2}(\mathbb{R})$ can span $M_{3\times 2}(\mathbb{R})$; (e) ten matrices in $M_3(\mathbb{R})$ can form a basis of $M_3(\mathbb{R})$; (f) three polynomials in $P_3(\mathbb{R})$ can span $P_3(\mathbb{R})$.

3. [Foundational]
Is $\{(1,1,0),\ (1,2,1),\ (0,1,3)\}$ a basis of $\mathbb{R}^3$? Row reduce the matrix whose columns are these vectors.

4. [Intermediate]
(Dimension depends on the scalars.) (a) Find a basis and the dimension of $\mathbb{C}^2$ regarded as a vector space over $\mathbb{R}$, and prove your set is a basis. (b) What is $\dim \mathbb{C}^2$ over $\mathbb{C}$? (c) Why is $\{(1,0),(i,0)\}$ linearly independent over $\mathbb{R}$ but linearly dependent over $\mathbb{C}$?

Part B — Bases of Subspaces Given by Constraints

5. [Foundational]
Let $W = \{(x, y, z) \in \mathbb{R}^3 : \ x = 2y\}$. Show that W is a subspace by exhibiting a basis, and state $\dim W$.

6. [Foundational]
Find a basis and the dimension of $W = \{A \in M_2(\mathbb{R}) : \ \operatorname{tr} A = 0\}$.

7. [Intermediate]
Find a basis and the dimension of $W = \{f \in P_3(\mathbb{R}) : \ f(0) = f(1) = 0\}$.

8. [Intermediate]
Find a basis and the dimension of $W = \{p \in P_3(\mathbb{R}) : \ p(0) = p(1),\ p(2) = 0\}$.

9. [Intermediate]
Find a basis and the dimension of $W = \{(x, y, z, w) \in \mathbb{R}^4 : \ x + y - z = 0, \ y + w = 0\}$.

10. [Intermediate]
Let $W = \{A \in M_2(\mathbb{C}) : \ a_{11} + i\, a_{12} = 0, \ a_{21} = (1 + i)\, a_{22}\}$, a subset of $M_2(\mathbb{C})$ over $\mathbb{C}$. Find a basis of W and $\dim W$.

11. [Intermediate]
Find a basis and the dimension of $W = \{A \in M_2(\mathbb{R}) : \ \operatorname{tr} A = 0 \text{ and } A = A^T\}$.

Part C — Extracting, Extending and Coordinates

12. [Foundational]
Find a linearly independent subset of $S = \{(1, 2, 1), \ (2, 4, 2), \ (1, 1, 0)\} \subseteq \mathbb{R}^3$ with the same span as S. State $\dim \operatorname{span} S$.

13. [Intermediate]

In $P_2(\mathbb{R})$ let $S = \{1 + x,\ 2 + 2x,\ x + x^2,\ 1 + 2x + x^2,\ x^2\}$. Find a linearly independent subset of S with the same span as S, express each remaining polynomial through it, and decide whether S spans $P_2(\mathbb{R})$.

14. [Intermediate]

Let $S = \{v_1, v_2, v_3\} = \{(1, 2, 3),\ (3, 2, 1),\ (1, 1, 1)\} \subseteq \mathbb{R}^3$. (a) Write the system that tests linear independence and row reduce it. (b) Decide whether S is linearly independent; if not, find an explicit relation. (c) Give a linearly independent subset of S that is a basis of $\operatorname{span} S$, and decide whether S spans $\mathbb{R}^3$.

15. [Intermediate]

Extend the linearly independent set $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \right\}$ to a basis of $M_2(\mathbb{R})$ by adding one of $E_{11}, E_{12}, E_{21}, E_{22}$. Justify your choice and say which of the four would *not* work.

16. [Intermediate]
Show that $\beta = \{1 - x,\ 2 + x^2,\ x + 2x^2\}$ is a basis of $P_2(\mathbb{R})$ and find the coordinate vector $[p]_\beta$ of $p(x) = 5 + 2x + 8x^2$.

17. [Intermediate]
Let $A_1 = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}$, $A_2 = \begin{pmatrix} i & -1 \\ 0 & i \end{pmatrix}$, $A_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ in $M_2(\mathbb{C})$. (a) Over $\mathbb{C}$, decide whether $\{A_1, A_2, A_3\}$ is linearly independent by row reducing, and give a basis of its span. (b) Regard $M_2(\mathbb{C})$ as a vector space over $\mathbb{R}$. Is $\{A_1, A_2, A_3\}$ linearly independent over $\mathbb{R}$? Prove your answer.

Part D — Proofs and Challenges

18. [Intermediate]
Let V be a vector space with $\dim V = n$ and let W be a subspace of V. Prove: (a) W is finite-dimensional with $\dim W \le n$; (b) if $\dim W = n$ then $W = V$.

19. [Intermediate]

Let W_1, W_2 be subspaces of a vector space V. (a) Prove that $W_1 \cap W_2$ is a subspace of V. (b) Prove that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

20. [Intermediate]

Let S_3 be the set of symmetric matrices in $M_3(\mathbb{R})$ ($A^T = A$) and K_3 the set of skew-symmetric matrices ($A^T = -A$). Find a basis and the dimension of each, proving that your sets are bases.

21. [Challenge]

Let $W = \{A \in M_2(\mathbb{C}) : \ A^* = A\}$ (here $A^* = \overline{A}^T$ is the conjugate transpose). (a) Show that W is a vector space over $\mathbb{R}$ and find a basis and $\dim_{\mathbb{R}} W$. (b) Show that W is *not* a subspace of $M_2(\mathbb{C})$ over $\mathbb{C}$.

22. [Challenge]

For which real numbers k is $\{(1,1,k),\ (1,k,1),\ (k,1,1)\}$ a basis of $\mathbb{R}^3$?

23. [Challenge]

Let V be a vector space with $\dim V = n$ and let $S \subseteq V$ have exactly n elements. Prove: (a) if S is linearly independent then S is a basis of V; (b) if S spans V then S is a basis of V. You may use: the Extension Lemma, the Dimension Bound, and that every basis of V has exactly n elements.

24. [Challenge]

Let S be a linearly independent subset of a vector space V and let $S' \subsetneq S$ be a proper subset. Prove that $\operatorname{span}(S') \neq \operatorname{span}(S)$.

25. [Challenge]

Let V be a vector space over $\mathbb{C}$ and $u, v, w \in V$. (a) Prove $\{u, v, w\}$ is linearly independent if and only if $\{u+v,\ u+w,\ v+w\}$ is. (b) Where did you use $2 \neq 0$? (c) Show that the statement fails over the field $\mathbb{F}_2 = \{0, 1\}$ (where $1+1=0$): give $u, v, w \in \mathbb{F}_2^3$.

26. [Intermediate]

Let $S = \{v_1, \ldots, v_n\}$ be a set of n distinct vectors in a vector space V. Prove that S is a basis of V if and only if every vector of V can be written *uniquely* as $a_1 v_1 + \cdots + a_n v_n$.

27. [Challenge]

Let $P(\mathbb{R})$ be the vector space of all real polynomials (of any degree). Prove that $P(\mathbb{R})$ is not finite-dimensional.